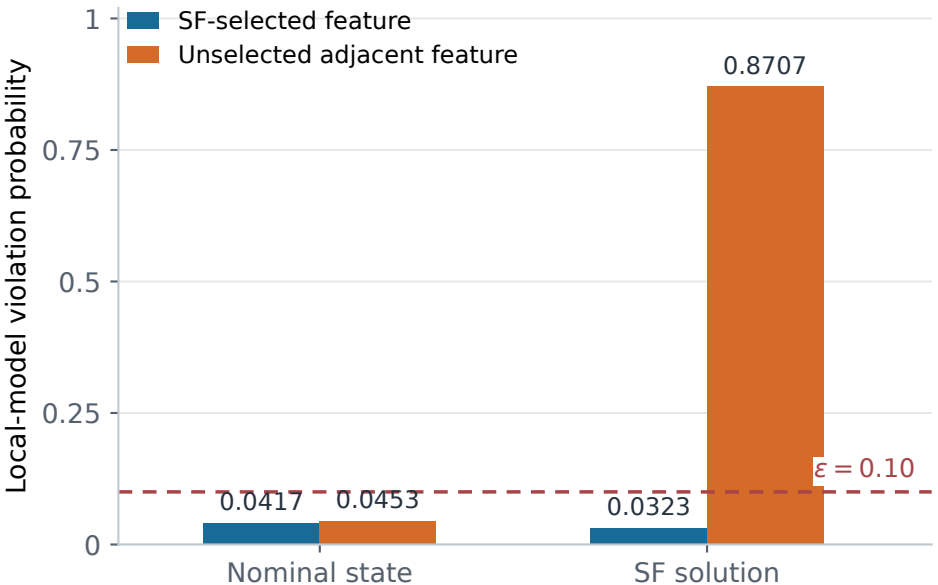
Representative feature-coverage failure

Robot width 126 mm | CC-SF, seed 5 | cycle 261 | 26.1–26.2 s

(a) Two fixed feature identities



(b) Constraint and geometry checks

Diagnostic	Value	Result
SF affine constraint at the SF solution	**+0.189 mm**	Passes $h \geq 0$
MF affine constraint at the same SF solution	**-1.573 quantile units**	Violates $\psi \geq 0$
Actual next pose signed PSDF	**-0.284 mm**	Overlap $\phi < 0$

MF: counterfactual, unrelaxed row; no reoptimization.

Nominal distance gap between these features: 0.01391 mm

Probabilities are frozen local-model estimates using nominal feature geometry and each state's planning covariance; they are not measured collision rates. The MF value evaluates the unrelaxed affine row at the SF solution, using the same nominal state, without MF reoptimization. Negative actual signed PSDF indicates geometric overlap. This is one case, not a guarantee that MF would prevent the event. Feature identities are held fixed from the nominal state to the SF solution.